

— As you arrive:

1. Make yourself a namecard with your preferred name (materials up front). (Please keep this and bring it to every subsequent lecture.)
2. If you haven't already done so, please take a look at the course syllabus (available on Canvas).

Lecture 1: Introduction, producer theory, and monopoly pricing

ECON 7510

Cornell University

Adam Harris

Slides draw upon lecture materials from Nikhil Agarwal (MIT).

Today's lecture

1. Introductions and course details
2. Industrial organization: Definition and history
3. Review of producer theory from ECON 6090
4. Monopoly pricing

Introduction to ECON 7510

Introductions

Getting to know me:

- Research: IO methods → questions in transportation economics and the economics of AI
- Preferred name: Adam

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Getting to know you:

1. Your preferred name
2. Your program and year
3. Something about your research interests and why you wanted to take the course / what you hope to get out of the course

Introductions

What experience do you come into the course with?

- ✓ Have you taken PhD-level microeconomics?
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- Have you solved dynamic programming problems numerically?

Course logistics

- Instructor office hours
 - Uris Hall 436.
 - By appointment. See syllabus for details and Calendly link.

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- Requirements
 - Assignments (30%):
 - » Five problem sets with empirical exercises and conceptual questions.
 - » Please turn in problem sets and code and output via Canvas.
 - » Coding in python, julia, or R strongly encouraged.
 - » You are encouraged to collaborate; groups can be as large as 3. Please list collaborators.
 - End-of-term assignment (in-class presentation or written) (30%): Either a review of a recent IO/structural paper or a proposal for a research project.
 - Class Participation (10%)
 - Final exam (30%): In-class November 12, 17, and 19. (You choose two of three.)

Course guidelines and expectations

Two principles:

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 - Lectures—questions and discussion encouraged.



Course guidelines and expectations

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- Choice of topics on final.

2. “Be curious, not judgmental.”

- Lectures—questions and discussion encouraged.
- We want to cultivate an environment where everyone can participate, ask questions, and learn
 - Please don't hold yourself back from participating!



Overview of Topics

Theme: Fundamental tools of industrial organization

- **Part 1:** Fundamental IO theory
 1. Producer theory
 2. Monopolies: Pricing; product quality/choice
 3. Price discrimination
 4. Competition: Static; dynamic
 5. Market entry
- **Part 2:** Computational building blocks
- **Part 3:** Estimating demand systems
- **Part 4:** Estimating single-agent dynamic discrete choice models

Industrial organization: Definition and history

Industrial organization

Question: What is industrial organization?

Industrial organization: Traditional definition

Industrial organization is the study of imperfectly competitive markets.

- How do firms behave, and what are the consequences for welfare?

(What determines the incentives of the firm?)

- Demand conditions
- Cost conditions – Production
- Competition

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- What determines demand, costs and competition conditions?
 - Investment, innovation
 - Market structure (entry, mergers, acquisitions)

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- What determines demand, costs and competition conditions?
 - Investment, innovation
 - Market structure (entry, mergers, acquisitions)
- Which policies, market rules should we adopt?
 - Antitrust policies
 - Regulation of concentrated industries
 - Market Design
 - Intellectual property

History of the field

1. Prior to 1980s: Structure-conduct-performance (SCP) empirical research
2. 1980s: Developments in theoretical IO
3. 1990s-present: New Empirical IO (NEIO)

Historical Perspective: Pre 1980

Schmalensee (1986): critical overview in Handbook of IO

- SCP: Structure - Conduct - Performance (Bain 1951, 1956)
- **Structure**: Inherent characteristics of the industry
 - Product type (demand)
 - Production technology (supply)
 - Number of firms (competitive environment)
 - Concentration (market power)
 - Entry barriers (fixed costs)
- **Conduct**: Firm behavior
 - Investment, innovation, pricing, advertising ...
- **Performance**: Positive or normative outcomes
 - Profits, welfare, entry, exit, R&D

SCP Paradigm

Cross-industry regression

- Study how structure $\implies$ conduct $\implies$ performance
- ✓ Difficult to observe conduct: focus on structure $\implies$ performance
- Cross-industry regression analysis:

$$\pi_i = \beta_0 + \beta_1 H_i + \beta_2 BE_i + \beta_3 (H_i \times BE_i) + \varepsilon_i$$

where i is industry (4 digit SIC)

$$\pi_i = \text{Profits}$$

$$H_i = \sum_j s_{ij}^2 \text{ is HHI (a measure of concentration)}$$

$$BE_i = \text{Entry barriers such as fixed costs, min efficient scale}$$

- Assume markets in long-run equilibrium

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- Assume markets in long-run equilibrium
- Do you have any concerns about this regression?

Issues with SCP

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Practical issues

1. Poor data
 - Accounting profits and costs are not economic profits/costs
 - Cross-industry analysis $\implies$ use data common to all industries (worst data)
2. Is 4-digit SIC the correct level of aggregation?
 - SIC code 2024 is ice cream and frozen desserts
 - SIC Code 7372 is prepackaged software

Policy relevance?

3. How do we measure consumer welfare?
 - Typical objective of antitrust authorities
4. Little to say about mechanisms (conduct)
 - Cannot guide public policy

Econometric issues

5. Endogeneity and reverse causality
 - Conduct may affect structure
 - Omitted variables

Example

Concentrated industries have few firms and high profits

- Two equally plausible explanations
 1. Market power has led to high prices
 2. Efficient firms may have driven away inefficient firms
- Matters for public policy

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- Two equally plausible explanations
 1. Market power has led to high prices
 2. Efficient firms may have driven away inefficient firms
- Matters for public policy
- Nevertheless, SCP literature was useful in establishing empirical regularities that spurred the theoretical literature to follow.

Theoretical IO: 1980s

(Part 1 of the course)

Approach and broad lessons (see Tirole's book)

- Game theoretic models of firm behavior
- Rich set of results explaining a wide range of phenomenon

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Approach and broad lessons (see Tirole's book)

- Game theoretic models of firm behavior
- Rich set of results explaining a wide range of phenomenon
- **A key takeaway:** Details of the specific market matter
 - ✓ Classic example: Investment could accommodate or deter entry depending on strategic effects (Fudenberg and Tirole, 1984)
 - Conundrum for public policy
- Tirole's Nobel lecture consequently calls for policies that pays attention to “specificities of particular industries”
 - ✓ Underlying theme: Heterogeneity across industries

New Empirical IO: 1990s - Present

(Parts 3-4 of the course)

Tirole's Nobel citation: “theoretical advances fundamentally affected the empirical IO literature”

- Theoretical literature offered:
 - Several explanations for similar observations
 - A basis for building empirical models
 - An understanding of important mechanisms
- Characteristics of NEIO work:
 - Focus on narrowly defined industries (heterogeneous effects)
 - Close attention to theory
 1. Deriving testable hypotheses
 2. Interpreting the data (structural modeling)
 - Relative to earlier empirical work, emphasis on
 1. Strategies (conduct)
 2. Quantifying welfare, economic costs, profits (performance)
 3. Counterfactual simulations (policy)

New Empirical IO Approach

- Phrase a question in terms of a counterfactual
 - ✓ What will happen if Union Pacific and Norfolk Southern merge?
 - Only data available is from a world where these are separate entities
- Approach: build the primitives of the model
 - Demand and supply
 - Interaction between railroads and customers
- Estimate a model: attention to data limitations and institutional details, e.g.
 - Prices negotiated between railroads and their customers might only be selectively observed.
- Simulate a counterfactual world and analyze outcomes of interest
 - Factual world: Competitive price setting and negotiations
 - Counterfactual: Joint price setting, considerations about pricing to competitors
 - ✓ Consumer welfare, profits, bargaining with upstream firms etc.

New Empirical IO: An example

Example: Use NEIO approach to study what would happen if Ford and GM merged.

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 - Consumer surplus
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With further extensions to the model, we can also analyze longer-run effects:

- Investment, innovation (of both merged firm and other firms)
- Entry of new firms

New Empirical IO

Questions: Am I an IO economist? Most of you don't plan to study imperfect competition. What are you doing in an IO course?

New Empirical IO → Structural empirical research

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Insight: The NEIO approach was developed to answer questions about imperfect competition, but it's actually much more widely applicable.

New Empirical IO → Structural empirical research

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Insight: The NEIO approach was developed to answer questions about imperfect competition, but it's actually much more widely applicable.

Examples (all recent job market papers):

- How do out-of-district campaign contributions distort the positions of Congressional candidates? (Waldfogel 2025)
- What would be the effects of reforming the structure of real estate agent commissions? (Kim 2025)
- Why do developing economies feature so many small firms? (Ramos and Sverdlin-Lisker 2022)
- How can policymakers effectively incentivize the development of combination therapies for treating cancer? (Dix and Lensman 2025)
- How can policymakers design mechanisms to incentivize conservation in a cost-effective way? (Aspelund and Russo 2025)

A Broader Definition?

- “Industrial Organization” may be too narrow to describe the modern IO field
- Perhaps “Market Organization” is better?
 - Analysis of market rules and structure
 - Behavior, incentives and payoffs of agents
 - Effects of policies and government interventions
- But even that might be too narrow. The field has become closely linked to a set of tools that are useful even in non-market contexts.
- Imperfect competition, antitrust, and regulation = “core IO”

Overview of Topics

Theme: Fundamental tools of industrial organization

- **Part 1:** Fundamental IO theory (**Core IO**)
 - (a.) Producer theory
 - (b.) Monopolies: Pricing; product quality/choice
 - (c.) Price discrimination
 - (d.) Competition: Static; dynamic
- **Parts 2-4:** Fundamental tools of structural estimation (**Broadly applicable empirical tools**)
 - 2. Computational building blocks
 - 3. Estimating demand systems
 - 4. Estimating models of single-agent dynamics

Producer theory: A review

Context

- Here, we'll briefly review some material from ECON 6090.
- From the ECON 6090 lecture notes:

Assumptions 3.1:

- (i) L commodities
- (ii) Production plan $y \in \mathbb{R}^L$
 - » Net input: good i such that $y_i < 0$
 - » Net output: good j such that $y_j > 0$
- (iii) Production possibility set, $Y \subseteq \mathbb{R}^L$ of feasible production plans
- (iv) Prices, $p \geq 0$, are unaffected by the activity of the firm.

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- To some extent, “core IO” is the study of 3.1.(iv) violations.
 - Today, we'll think about what happens when this assumption holds and when it does not.

Technological feasibility

Assumptions 3.1:

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- (iv) Prices, $p \geq 0$, are unaffected by the activity of the firm.

Assumptions 3.2:

- (i) Y is nonempty, closed and (strictly) convex.
- (ii) Free disposal: If $y \in Y$ and $y' \leq y$, then $y' \in Y$.

Single-output case: $f(z) = \max_q q$ s.t. $(-z, q) \in Y$

Efficiency

Definition: A production plan $y \in Y$ is *efficient* if there does not exist a $y' \in Y$ such that $y' \geq y$ and $y'_i > y_i$ for some i .

Profit maximization

General case:

$$\pi(p) \equiv \max_y p \cdot y \text{ subject to } y \in Y$$

Single-output case:

$$\pi(p, w) \equiv \max_{z \in \mathbb{R}^{L-1}} pf(z) - w \cdot z$$

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FOC:

$$\underbrace{p \nabla f(z)}_{\text{MRP}} = w \Rightarrow \frac{f_i(z)}{\underbrace{f_{i'}(z)}_{\text{MRTS}}} = \frac{w_i}{w_{i'}}$$

Profit maximization implies cost minimization

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Profit maximization implies cost minimization

$$\begin{aligned}\pi(p, w) &\equiv \max_{z \in \mathbb{R}^{L-1}} pf(z) - w \cdot z \\&= \max_q \left[\max_{z \in \mathbb{R}^{L-1}} pq - w \cdot z \text{ s.t. } f(z) = q \right] \\&= \max_q pq - \left[\underbrace{\min_{z \in \mathbb{R}^{L-1}} w \cdot z \text{ s.t. } f(z) = q}_{\text{CMP}} \right] \\&= \max_q pq - C(w, q)\end{aligned}$$

Profit maximization (with product market power)

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Profit maximization implies cost minimization (with product market power)

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Quantity choice under perfect competition

$$\max_q pq - C(q)$$

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FOC:

$$p = C'(q)$$

Price equals marginal cost. Zero profit on the marginal unit.

Monopoly pricing

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$$[p(q^m) + p'(q^m)q^m] = C'(q^m)$$

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Marginal revenue equals marginal cost.

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Marginal revenue equals marginal cost.

$$\begin{aligned}\Rightarrow p(q^m) &= C'(q^m) - \underbrace{p'(q^m)q^m}_{<0} \\ &> C'(q^m)\end{aligned}$$

Positive profit on the marginal unit.

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Positive profit on the marginal unit. *How much profit?*

Monopoly pricing

Equivalently, price choice: (Notation: $D = p^{-1}$.)

$$\max_p pD(p) - C(D(p))$$

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FOC:

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$$p^m - C'(D(p^m)) = -\frac{D(p^m)}{D'(p^m)}$$

$$\frac{p^m - C'(D(p^m))}{p^m} = -\frac{D(p^m)}{D'(p^m)p^m}$$

“Lerner Index”: $L = -\frac{1}{\epsilon}$

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“Lerner Index”: $L = -\frac{1}{\epsilon}$

Question: What does this imply about perfectly elastic demand? Unit elastic demand?

Properties of monopoly pricing

- What point on the demand curve does monopolist choose?

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$$p^m = \left(\frac{1 + \epsilon}{\epsilon} \right) C'$$
$$p^m > 0 \Leftrightarrow \frac{1 + \epsilon}{\epsilon} > 0$$
$$\Leftrightarrow \epsilon < -1$$

Elastic part of the demand curve.

→ As long as demand is inelastic, $\frac{\partial \pi}{\partial p} > 0$, so increase price (i.e., decrease quantity) until you get to an elastic part of the demand curve.

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- Inefficiency? Yes, any deviation from $p = C'$ means quantity is inefficient.
- p^m is weakly increasing in marginal cost.

Proof: p^m is weakly increasing in marginal cost.

- Suppose $C_2'(q) > C_1'(q)$ for all $q > 0$.
- Let (p_1, q_1) and (p_2, q_2) denote the corresponding monopoly prices and quantities.
 - **Key idea:** Both (p_1, q_1) and (p_2, q_2) are points on the demand curve, so both feasible for both monopolists.

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- Combining the two:

$$[C_2(q_1) - C_1(q_1)] - [C_2(q_2) - C_1(q_2)] \geq 0$$

Proof: p^m is weakly increasing in marginal cost.

- Suppose $C_2'(q) > C_1'(q)$ for all $q > 0$.
- Let (p_1, q_1) and (p_2, q_2) denote the corresponding monopoly prices and quantities.
 - **Key idea:** Both (p_1, q_1) and (p_2, q_2) are points on the demand curve, so both feasible for both monopolists.

$$p_1 q_1 - C_1(q_1) \geq p_2 q_2 - C_1(q_2)$$

$$p_2 q_2 - C_2(q_2) \geq p_1 q_1 - C_2(q_1)$$

- Combining the two:

$$[C_2(q_1) - C_1(q_1)] - [C_2(q_2) - C_1(q_2)] \geq 0$$

which implies

$$\int_{q_2}^{q_1} \underbrace{[C_2'(x) - C_1'(x)]}_{>0 \ \forall x} dx \geq 0$$

so $q_1 \geq q_2$, which means $p_1 \leq p_2$.

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 - Dynamic considerations: Reputation, resale market for durable goods, learning by doing, etc.
 - Multiproduct firm.

Next time

More monopoly:

1. Multi-product pricing
2. Durable goods
3. Quality choice